

Design and Fabrication of Hybrid Functional Identities for Mechanical Elements

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Abstract

My PhD research explores the simultaneous integration of mechanical and electrical functionalities in mechanical components such as gears, linkages, and springs, which I define as “hybrid functional identities.” The focus is on transforming these components into non-intrusive sensors and active elements that maintain structural integrity while providing electrical capabilities like sensing, energy harvesting, and communication. I establish a framework for hybrid functional identities by examining common mechanical elements and their associated motions—rotational, linear, and reciprocal—along with force-based interactions like stretching, compression, and torsion. This analysis identifies essential electrical functionalities that complement these mechanical behaviors. Building on this foundation, I investigate modular mechanical building blocks that support diverse mechanical and electrical interaction primitives using a unified geometric structure. Ultimately, I aim to create an interconnected system where hybrid mechanical-electrical components function autonomously and communicate through an embedded wireless network.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

metamaterial mechanisms, reconfigurable antenna, wireless communication and sensing

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Forward

I am a fourth-year PhD student in Mechanical Engineering at MIT, advised by Professor Stefanie Mueller. My research focuses on the fabrication of interactive mechanical components that incorporate sensing, energy harvesting, and wireless communication capabilities, enhancing them with hybrid functionalities. My work has

been published in prominent HCI venues, such as CHI, UIST, and SCF, and is supported by the Bahrain Crown Prince International Scholarship Program. I have successfully passed my Mechanical Engineering Qualification Examination, and expect to complete all required coursework this year. My projected graduation date is Summer 2027. After graduation, I plan to pursue academic positions focused on developing intelligent multi functional physical systems. I aim to explore how physical components can be augmented with capabilities that are traditionally realized through external components, allowing them to function independently and cohesively.

1 Context and Motivation

Many mechanical systems in robotics and adjacent fields rely on electrical components to provide both input and output for mechanical motion. For example, encoders are used in rotational mechanisms as sensors that output angular position and velocity feedback, while motors act as sources of motion by translating electrical signals into physical movement.

The addition of sensors, or other electrical components such as energy sources, can very often add additional constraints when engineers design mechanical assemblies. For example, routing wires, or housing a sensor might prompt change to a whole assembly's design. Researchers within HCI have addressed the challenge of integrating electronic components into 3D printed models by creating design tools that automate this process. For instance, some researchers have developed tools that enable users to embed electronic parts within a 3D printed model, complete with electrical routing paths [4] [9], while others have focused on embedding sensing capabilities into input widgets like sliders and buttons that interface with screens [16] [17] [8] [18] [2] [12] [14], or into mechanical meta-material cell structures [3] [20] and deformable lattices [15]. While these studies have made significant strides in developing design tools and fabrication methods for integrating electronic dimensions into physical structures, their primary focus has been on enhancing general interactivity within 3D models rather than addressing the specific electrical functionality needs associated with mechanical components, like requiring an external encoder to measure the velocity of a wheel. The work that most closely aligns with my dissertation is SteelSense[19], which introduces the concept of functional hybrid electrical identity for mechanical elements. Steelsense's approach does have some limitations concerning the effectiveness of sensor placement and mechanical functionality within assemblies. For example, in rotational components like the rack/pinion gear set, there are concerns about potential wire tangling on moving parts, and the placement of conductive areas may present challenges for certain fabrication methods.

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In my research, I introduce the concept of unifying mechanical and electrical functionalities by transforming mechanical components into sensors, for example. Rather than adding or generating separate sensors or electrical components into mechanical assemblies, I focus on non-intrusive methods that preserve the structural integrity of the components while enabling them to also perform as effective electrical elements. This approach maintains the original mechanical function while embedding electrical capabilities directly into the structure.

To make this integration scalable and applicable to a wide range of structures, I investigate the fundamental building blocks corresponding to different types of fundamental mechanical motions—rotational, linear, reciprocal, and combined. I then extend these building blocks to not only be complimentary to a machine’s motion but also to basic human physical interaction affordances which include stretching and compression, bending, rotation/torsion, and shearing of tangibles. I define the dimension of electrical interactivity or identity that is augmented to these mechanical components through three primary functions: sensing (measurement and detection of events), energy harvesting (generation of a power source), and wireless communication (building of a communication network).

The primary contributions of this dissertation lie in my approach to integrating mechanical and electrical functionalities within interactive systems seamlessly. First, this dissertation offers a comprehensive framework for understanding how mechanical components can inherently possess electrical capabilities without compromising their structural integrity or primary mechanical functions. Second, it presents empirical findings that validate the effectiveness of these integrated systems in real-world applications. Finally, this work contributes to the broader field by providing design guidelines and methodologies for engineers seeking to create hybrid mechanical-electrical structures. Through my dissertation I hope to advance the state of mechanical-electrical design, by demonstrating how traditional mechanical components can be re-envisioned as multi-functional entities.

2 RESEARCH QUESTIONS

Throughout my PhD, I address the challenges within the integration of mechanical and electrical systems through a series of research questions organized into four main sub-themes:

1. Electrical and Mechanical Dimensions

- RQ1-1: What basic mechanisms are predominantly used in mechanical assembly design?
- RQ1-2: What electrical functionality dimensions are most beneficial for integration into mechanisms?
- RQ1-3: What types of motion or mechanisms are complimentary to these electrical dimensions?

2. Methods for Seamless Hybrid-Identity Integration

- RQ2-1: How can we seamlessly integrate these electrical dimensions into the structure without compromising the mechanical identity and performance?

3. Creation of Building Blocks and Integrated Networks

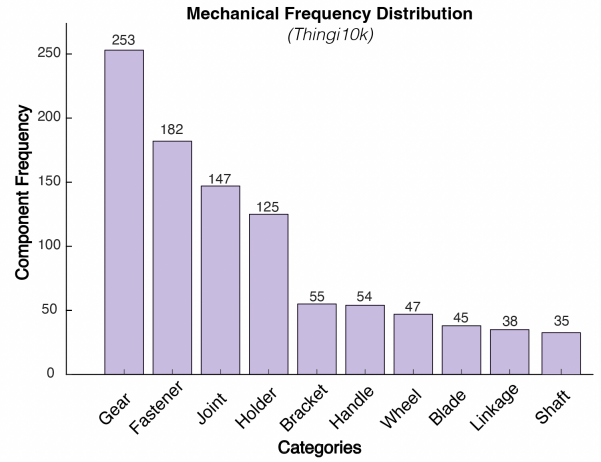


Figure 1: Distribution of mechanical component frequencies within the Thingi10k dataset for 3D Printed models. The graph illustrates the prevalence of various rotating mechanisms such as gears, joints and wheels.

- RQ3-1: How can we design these elements to be modular enough to serve as building blocks that can be translated into any base motion?
 - RQ3-2: How can we enable these interactive building blocks to communicate with one another, forming a cohesive network?
4. Evaluation of Performance
- RQ4: How does the performance of these hybrid mechanical components compare to off-the-shelf electronic elements?

3 Research Methods

I use a variety of research methods to address design and technical challenges. This includes surveying mechanical elements frequently fabricated in maker communities and developing user interface tools informed by these insights, such as Solidworks plug-ins (C#) and sensing tool UIs (Java, C++, Python), as demonstrated in [1]. I primarily rely on rapid prototyping, using multi-material 3D printing and laser cutting. Additionally, I conduct performance benchmarking through technical evaluations and analyze data using custom Matlab and Python scripts to ensure a rigorous, data-driven approach.

4 Findings to Date

4.1 RQ1-1: What basic mechanisms are predominantly used in mechanical assembly design? [1] [13] [ONGOING]

In MechSense [1], I conducted a formative study of the Thingi10k repository on Thingiverse, which features 10,000 objects, including duplicates of the same component, created by the maker community. The study aimed to identify common mechanical components used by hobbyists with varying levels of design expertise. Mechanical

engineers were tasked with categorizing unique (non-duplicated) models in Thingi10k as either mechanical or non-mechanical components. After every 1,000 components, participants reviewed their classifications to resolve disagreements and refine the criteria for what constitutes a mechanical component (e.g., whether a “battery holder” qualifies). Ultimately, 981 unique part files were unanimously identified as mechanical.

The study revealed that rotational components—such as gears, joints, wheels, and linkages—are the most prevalent in mechanical assemblies within maker projects [1]. These findings directly informed the first project of my dissertation, which focuses on integrating electrical functionality into rotational mechanical elements.

I also plan to expand this formative study by exploring online repositories with a stronger mechanical focus, such as McMaster-Carr.

However, relying solely on rotational components and listing top mechanisms provides a limited perspective for creating a comprehensive framework for hybrid-functionality mechanical design. To address this, I expanded my research to two additional dimensions: (1) the type of motion and (2) force-based mechanical interaction primitives. Dimension 2 is crucial because, while gears, for example, support multiple motion types—linear, rotational, and reciprocal—they fall short with interactions like stretching, compression, and torsion, which are essential for broader applications. Recognizing this, I explored mechanical elements capable of supporting these interaction primitives, focusing on cantilever beams, as in Acceloprint [13], which enable bending, torsion, compression, and shearing. Building on this, I am now exploring modular meta-material structures. These structures combine multiple interaction primitives into cohesive, modular building blocks, potentially replacing complex assemblies [5] [6]. This approach enables a versatile, integrated design for mechanical systems.

In summary, my research began by identifying common motion types, such as rotational elements [1], then expanded to mechanisms supporting different motions and affordances [13]. Now, I focus on integrating these dimensions into modular structures that serve as building blocks for more complex mechanical systems.

4.2 RQ1-2: What electrical functionality dimensions are most beneficial for integration into mechanisms?

Electrical components can be categorized as either passive, such as sensors, or active, like actuators and energy sources. Some components, like antennas, serve both roles by transmitting (active) and receiving (passive) signals. My work begins with passive sensors and progresses to active energy sources, ultimately aiming to create a seamless communication ecosystem among these hybrid-functionality elements through RF communication (antenna).

These electrical functionalities rely on essential building blocks for electro-mechanical systems: sensors provide feedback and measurement, energy sources enable actuation, and RF communication, such as WiFi, facilitates transmission and reception over distances.

4.3 RQ1-3: What types of motion or mechanisms are complimentary to these electrical dimensions? [1] [13] [ONGOING]

To address this question, I began by examining the primary functions of mechanical elements and highlighting the role of electrical components in defining their mechanical identity. For example, rotation—often achieved through gears or wheels—typically necessitates position or speed sensing, which is why encoders are commonly used [10] [11], hence a sensing electrical dimension [1]. Next, I explore beams, which usually serve as structural building blocks, but can also act as actuated elements such as flexible end effectors that bend. To enhance their capabilities, I integrate accelerometers into these beams for multi-axis sensing [13].

I also delve into reciprocating motion in my ongoing work, where sliding or separating mechanical elements produce linear motion, a fundamental principle in triboelectric energy harvesting, adding an energy generation functional dimension.

4.4 RQ2-1: How can we seamlessly integrate these electrical dimensions into the structure without compromising the mechanical identity and performance? [1] [13] [ONGOING]

In MechSense [1], I addressed the integration of electrical sensing within rotational mechanisms. Given the non-contact nature of rotations and the challenge of wire entanglement between stator and rotor, I chose capacitive sensing, a non-contact based sensing method. I then developed a floating capacitive encoder embedded on the surface of rotating components, which avoids compromising structural integrity by positioning the sensor away from areas of direct mechanical stress. The use of multi-material 3D printing allowed for the creation of these components in a single fabrication step, eliminating the need for post-assembly. This design strategy of multi-material integration informs my ongoing work on seamless geometric integration for other types of mechanical elements.

4.5 RQ3-1: How can we design these elements to be modular enough to serve as building blocks that can be translated into any base motion? [ONGOING] [PROPOSED]

The modular design of placing conductive material on the surfaces of rotors and stators to perform as a capacitive pair, introduced through MechSense, is adaptable to any rotating element. This is demonstrated by MechSense’s applications from a linkage-based lamp, to a gear train, and a wheel Figure-2. My ongoing work focuses on using metamaterial structures, which have been demonstrated through research in material science and HCI as being promising building block candidates to enhance this desired modularity for mechanism design—especially since they are compatible across a variety of mechanical and electrical dimensions.

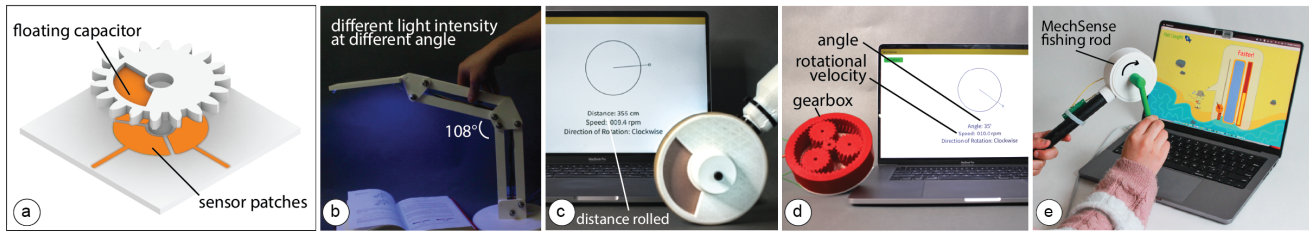


Figure 2: (a) MechSense enables 3D printed mechanisms to detect their rotational direction, speed, and angular position by utilizing a floating capacitor on the moving component and sensor patches on the stationary part. (b) A smart lamp with linkage-based movement that adjusts light intensity when the top linkage bars are shifted, (c) a measuring wheel capable of determining surface dimensions, (d) a planetary gearbox that can sense its own operational state, and (e) a fishing rod-shaped controller that interacts with a 2D game.

5 EXPECTED NEXT STEPS

5.1 RQ3-2: How can we enable these interactive building blocks to communicate with one another, forming a cohesive network? [PROPOSED]

My vision goes beyond embedding hybrid functionalities into individual components. I aim to create a network of interconnected mechanical elements that communicate and interact, much like the interdependent systems in the human body. For instance, if a shaft bends or distorts, it could signal another mechanical part to adjust and correct the issue, or even trigger external elements to join the system and assist in error correction. This will be achieved by integrating RF communication capabilities through antenna structures embedded within 3D models. Research within HCI community, such as Printed Wifi [7], have already demonstrated the potential for mechanisms to wirelessly transmit data.

5.2 RQ4: How does the performance of these hybrid mechanical components compare to off-the-shelf electronic elements? [1] [ONGOING]

In the case of MechSense [1], I evaluated the performance of my 3D-printed encoders against off-the-shelf alternatives. The results were promising, with performance levels deemed acceptable for prototyping applications. A detailed technical evaluation will be conducted for each additional function, such as energy harvesting and RF communication, to ensure reliability and performance parity with commercial electronic components.

6 Research Community Activities

I have actively contributed to mentoring and collaboration within research communities. I have mentored 11 undergraduate students from MIT, Dartmouth, and Cornell on HCI research, many of whom are co-authors on my research papers. I have also collaborated with researchers from two universities outside the U.S., TU Delft in the Netherlands and Aarhus University in Denmark. In 2023, I launched the MIT-Bahrain Global Teaching Lab for Science and Technology—the first ever university-level teaching program between MIT and

a Gulf country. I have proposed two workshops on robotics and wearable technologies through the program, and this year, will be teaching the first university-level HCI fabrication course in Bahrain on actuated experience design. I have developed the course curriculum from scratch alongside my MIT collaborator, Yunyi Zhu.

7 Conclusion

To summarize, my research contributes: (i) a novel framework for hybrid functional identities in mechanical components, enabling them to serve dual roles as mechanical systems and electrical elements, such as sensors, energy harvesters, and communicators; (ii) design techniques that leverage the intrinsic mechanical properties of components to incorporate electrical functionalities, ensuring non-intrusive integration without compromising structural integrity; (iii) tools that simplify the process for end-users to create and control hybrid mechanical-electrical assemblies in applications ranging from robotics to smart interactive devices; (iv) demonstration of fabrication techniques that enable the seamless production of these hybrid components, embedding electrical capabilities directly into the mechanical fabrication process; and (v) a diverse range of applications that benefit from the enhanced capabilities of interconnected mechanical-electrical systems, including robotics and smart-home appliances. In addition to these ongoing research efforts, I have outlined future projects aimed at further demonstrating the potential of hybrid functional identities for energy harvesting and establishing a wireless network ecosystem. In my prior projects, I investigated the incorporation of sensors into mechanical components to enable feedback, and the next step involves developing energy harvesters and RF communication capabilities to facilitate seamless communication within an interconnected system. The development of hybrid functional identities represents a significant advancement in how we design mechanical components, opening new pathways for interaction and functionality. This ongoing exploration positions my research at the forefront of how we reshape our understanding of mechanical systems and their applications in everyday life. By focusing on the seamless integration of mechanical and electrical functionalities, I aim to pave the way for future machines that are not only mechanically capable, but also intrinsically intelligent and responsive to their environments.

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